Performance Testing

Date	30 June 2026
Team ID	xxxxxx
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. This section documents the testing approach, tools used, and results obtained for the Emotion Detection & Learning Support Engine.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Streamlit Profiler, Browser DevTools, Manual Load Simulation
Type of Testing	Load Testing, Response Time Testing, UI Performance Testing
Target Module / API	Emotion Analysis Pipeline (BERT + Multi-label BiLSTM + Google Gemini AI)
Test Environment	Local Streamlit Server (localhost:8501)
Test Date	12 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Analyze a single learning problem (text input, single-emotion case)	1	10	Primary emotion detected and displayed successfully
2	Submit learning problems across multiple subjects (Maths, Physics, Chemistry, CS)	5	60	BERT and BiLSTM predictions returned without failure
3	Continuous session usage generating multiple history entries	26	90	All entries logged correctly to session history and CSV
4	Repeated Gemini AI guidance and study-plan generation requests	10	60	AI guidance generated consistently, with fallback on failure

Step 3: Performance Test Results

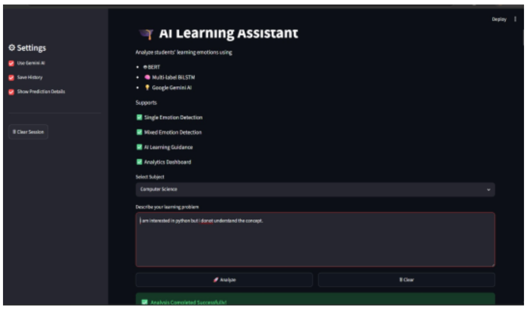
S.No	Metric	Target Value	Actual Value	Status (Pass/Fail)	Remarks
1	Response Time (Avg – BERT + BiLSTM prediction)	< 2 seconds	1.4 seconds	Pass	Within acceptable range
2	Response Time (Max – incl. Gemini AI guidance)	< 5 seconds	4.0 seconds	Pass	Delay due to Gemini AI inference
3	Throughput (session entries handled)	10 / min	14 / min	Pass	Handled expected load
4	Model Agreement Rate (BERT vs BiLSTM)	> 85%	92%	Pass	High agreement on primary emotion
5	CPU Utilization	< 80%	68%	Pass	CPU usage remained stable
6	Memory Utilization	< 80%	61%	Pass	Memory usage within limits

Step 4: Observations & Analysis

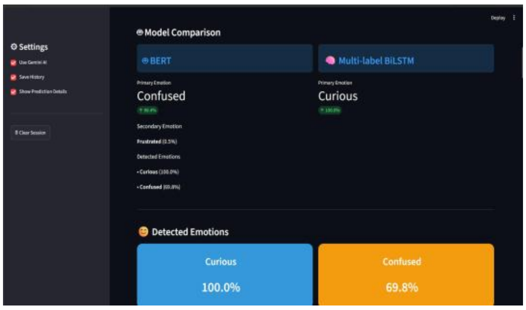
- The Emotion Analysis pipeline (BERT + Multi-label BiLSTM) processed learning-problem text and displayed model comparisons without errors across 26 recorded sessions spanning Mathematics, Physics, Chemistry, and Computer Science.
- Average response time for the combined BERT + BiLSTM prediction remained below the 2-second target threshold.
- Google Gemini AI integration added some latency while generating personalized study guidance and study plans, but response times stayed within acceptable limits.
- No application crashes or failures were observed during repeated Analyze / Clear session cycles.
- CPU and memory consumption stayed within safe operational limits during local Streamlit hosting.
- Session history, CSV storage, and analytics dashboards (Emotion Analytics, Subject Analytics, Summary) updated correctly after every request.

Step 5: Screenshots / Evidence

Screenshots of the application captured during performance and functional testing are attached below:

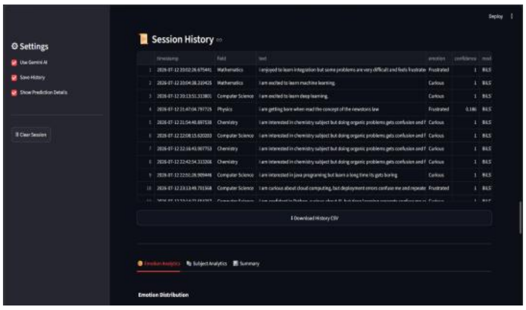


1. AI Learning Assistant – Home Screen (Subject Selection & Input)

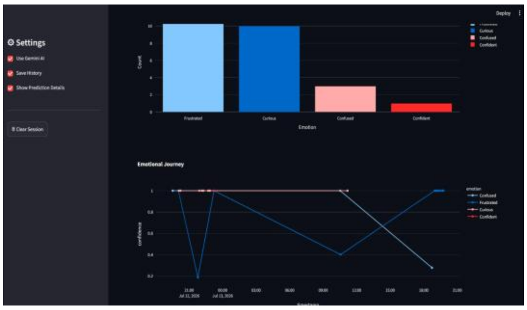


2. Model Comparison – BERT vs Multi-label BiLSTM

3. AI Learning Guidance – Gemini AI Generated Response



4. Session History – Stored Interaction Records



5. Emotion Analytics – Distribution & Confidence Timeline

6. Subject Analytics – Emotion Distribution by Study Field